

Online Processing System for Molten Salt Reactors

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INTRODUCTION

The department of energy predicts domestic electricity demand could more than double by 2050. However, achieving net-zero carbon energy production would require roughly an addition 700 to 900 GW of clean generation capacity [1]. For reference, the current generation capacity of the United States is 1300 GW [2]. As the energy demands surge over the next 30 years, advanced nuclear reactors will be instrumental in supplanting fossil fuels and creating a net-zero carbon future.

According to the 2018 R&D outlook published by the Gen IV International Forum (GIF), there are six promising generation IV reactor designs, namely: (i) sodium-cooled fast reactors (SFRs), (ii) very high temperature reactors (VHTRs), (iii) gas-cooled fast reactors (GFRs), (iv) molten salt reactors (MSRs), (v) lead-cooled fast reactors (LFRs), and (vi) supercritical water-cooled reactors (SCWRs). Of these advanced reactors, MSRs are of particular interest for their high thermal efficiency, breeding capabilities, and versatile fuel types.

One of the important R&D challenges laid out by the GIF is on-site fuel processing [3]. MSRs are particularly well suited to on-site fuel processing due to their circulating, liquid fuel. In this paper, we discuss removal methods for tritium and neutron poisons from the primary loop of MSRs. Removing tritium is crucial for the lifetime of the reactor as tritium at high temperatures, like those in MSRs, is highly corrosive threatening the integrity of structural materials [4]. Additionally, the removal of relevant neutron poisons, like ^{135}Xe and ^{149}Sm , is crucial to an improved neutron economy.

To address this demand for on-site fuel processing, we propose a multi-stage removal system. These removal mechanisms are constituted by mechanical and chemical separation processes on either side of the heat exchanger (HX). The mechanical separation processes are upstream of the HX and include a helium bubble generator, gas separator [5], and graphene absorption columns. The chemical processes are down-stream from the HX, due to the removal of energy caused by the separation, and the bulk of available research into electrochemical extraction being conducted at temperatures less than that of the reactor core operating temperature [6].

By coupling the aforementioned components, we will design an online processing system integrated into the primary loop of MSRs. With this system, MSRs will have less corrosion and an improved neutron economy, bringing them closer to fruition.

THEORY

This project is being completed in conjunction with our senior design course at the University of Illinois at Urbana-Champaign. Unfortunately, the senior design course begins

January 22, after this summary is due. Therefore, the system components and design we will discuss are preliminary and should be considered with such discretion in mind. For the final submission, we will have a more rigorous description for each component, justification of its inclusion, and potentially more or different components.

Components

The basis for this paper is designing an online fuel processing system for MSRs. Therefore, we will attach our reprocessing system to a model of the Molten Salt Reactor Experiment [7].

The first component in our system is a helium bubble generator [5], injecting a stream of helium bubbles through the flowing salt. These bubbles flow with the salt, allowing gaseous fission products to diffuse from the salt into the bubbles. These bubbles are then removed by the next component, the bubble separator. To remove the tritium from the system, we propose an absorption column, which readily absorbs tritium as the salt flows past [8]. This absorption column would be made of graphene due to the storage capacity of graphene compared to other forms of carbon – graphene has a tritium storage capacity of 0.9 %_{wt} while the values for other, more exotic, more expensive forms of carbon are: 0.82 %_{wt} for TC-3, 0.72 %_{wt} for Maxsorb, 0.66 %_{wt} for MOF-177, and 0.62 %_{wt} for AX-21. The absorption column leverages the high storage capacity and hydrogen affinity of graphene to passively remove graphene from the primary loop. The final component of our system is an aluminum coated tungsten electrode be used after the heat exchanger, which will separate ^{149}Sm by electro-reduction of Sm(III) [6],

Computational Models

MSRs are notoriously difficult to model with high accuracy due to their flowing fuel and so, in this report, we utilize multiple computational tools to model the various necessary physics present in MSRs.

To begin, we will utilize SaltProc [9], coupled with OpenMC [10], for depletion analysis of our core. SaltProc leverages OpenMC's depletion solver to generate core inlet and outlet concentrations of all isotopes. SaltProc uniquely offers neutron poison removal capabilities, which we will use to accurately model the poison removal of our removal system.

We will utilize Moltres [11] for in-core transient analysis. Moltres is a multiphysics simulation tool, leveraging a coupled neutron diffusion and Computational Fluid Dynamics (CFD) solver for accurately modeling delayed neutron precursor (DNP) drift and strong temperature feedback. We will utilize Moltres to simulate accident scenarios, in which neutron poisons are reintroduced to the core at the inlet. To accu-

rately model this reintroduction, we introduce new capabilities to Moltres: spatially resolved poison tracking. This feature uses the Finite Element Method (FEM) solver in Moltres to solve Equation 1, with all material properties generated with OpenMC's group microscopic cross-section generation and depletion features. Subscripts i and p denote the isotope of interest and parent isotopes, respectively. Each of the remaining terms are defined in Table I.

$$\begin{aligned} \frac{\partial C_i}{\partial t} = & \gamma_i \sum_{g=1}^G \Sigma_{f,g} \Phi_g + \sum_p^{P_t} \sum_{g=1}^G \sigma_{p \rightarrow i} C_p \Phi_g \\ & + \sum_p^{P_d} \lambda_{p \rightarrow i} C_p - \sum_{g=1}^G \sigma_{a,g,i} C_i \Phi_g - \lambda_i C_i - \mathbf{u} \cdot \nabla C_i \quad (1) \end{aligned}$$

TABLE I. Nomenclature in Poison Tracking Equation

Term	Definition
C	Concentration
Φ_g	Scalar neutron flux of group g
G	Number of neutron energy groups
P_d	Number of decay parents
P_t	Number of transmute parents
γ	Fission Yield
$\Sigma_{f,g}$	Macroscopic fission xs of group g
$\sigma_{p \rightarrow i}$	Microscopic absorption xs of parent p to i
$\lambda_{p \rightarrow i}$	Decay constant from parent p to i
$\sigma_{a,g,i}$	Microscopic absorption xs of i for group g
λ_i	Decay constant of i
$\mathbf{u}$	Fluid velocity

RESULTS

Although our results are not complete, we aim to present the efficacy of an online processing system. This system will remove gaseous fission products, neutron poisons ^{135}Xe and ^{149}Sm , and numerous other fission products, such as tritium. For each component, we will conduct an economic and structural analysis to determine the integration feasibility into future MSR designs.

For computational methods, we will conduct steady-state neutronic analysis with OpenMC to ensure a working and safe reactor is achievable. If we have the time, we will also conduct a transient neutronic analysis with Moltres to investigate the implications of system failure. Finally, we will conduct a 1-D Control Volume (CV) analysis ensuring the feasibility of our system.

CONCLUSIONS

As we do not yet have results, we have no resolute conclusions. For the final paper submission, we will determine the optimal component layout to maximize the lifetime of reactor components of interest, namely: the heat exchanger, primary salt pumps, and piping. In the component layout, we will determine the order of the components and which components would be included. Ideally, the components we choose to implement directly after the core outlet will purify the contaminants in the primary loop minimizing the long-term operational costs. Alternatively, we will show the proposed components are not economically or technically feasible given the current technology.

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